Name : Hamid Reza Ghahramani Date :6 OCT 2026

A data sheet titled **"Deep Brain Stimulation Targets in Epilepsy: Systematic Review and Meta-analysis of Anterior Thalamic Nuclei"** has been provided.

1. clean the data corresponding to the **Epilepsy_Type** and **Quality** variables. Then, import the cleaned dataset into Stata. (Screenshots should be taken throughout the process)

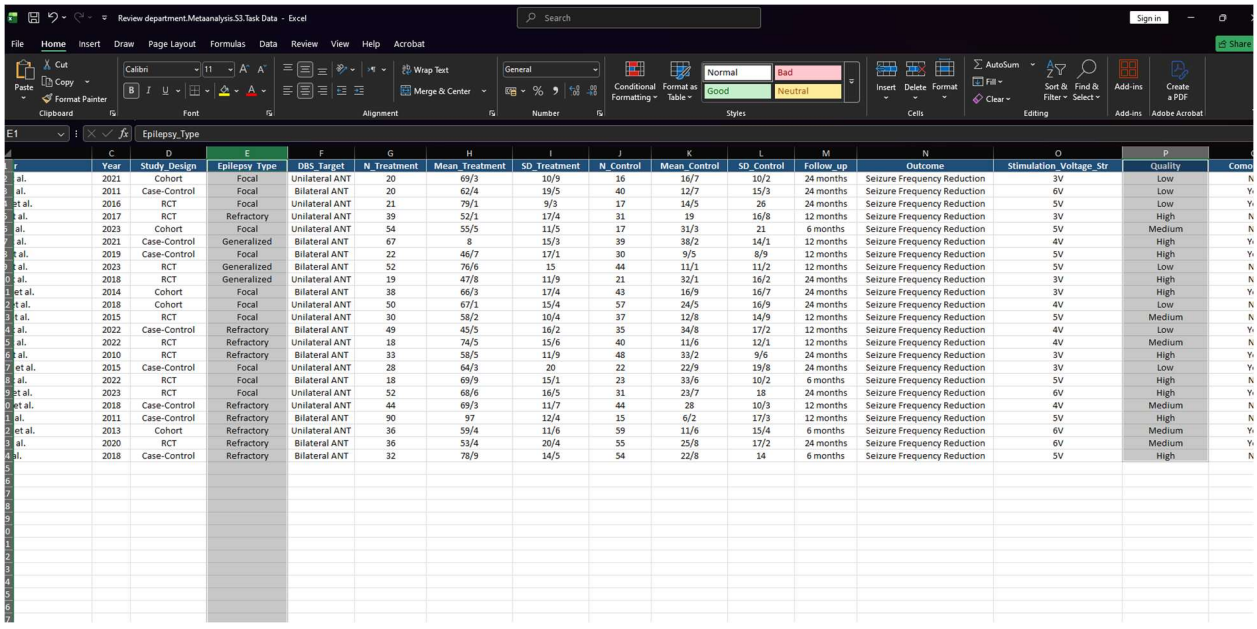
2. Calculate the effect sizes and generate a forest plot. Then, interpret the forest plot by addressing the following points:

1. Explain what the **overall effect** means.
2. Report the **number of studies with non-significant results**.
3. Based on visual inspection, determine whether there appears to be **heterogeneity** among the studies and, if so, provide an approximate assessment of its extent. (low, moderate, high)
4. Identify and report the **number of data points that you suspect may be outliers**.

Design and format the forest plot according to your own preferences.

(Screenshots should be taken throughout the process)

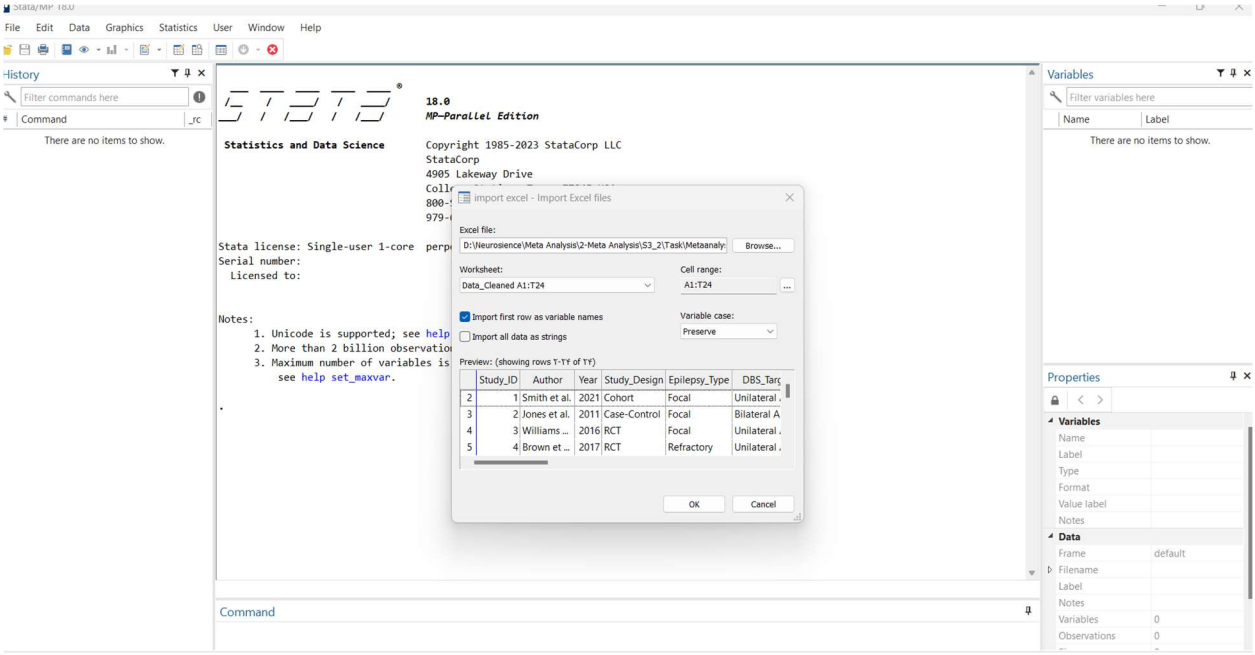
1.



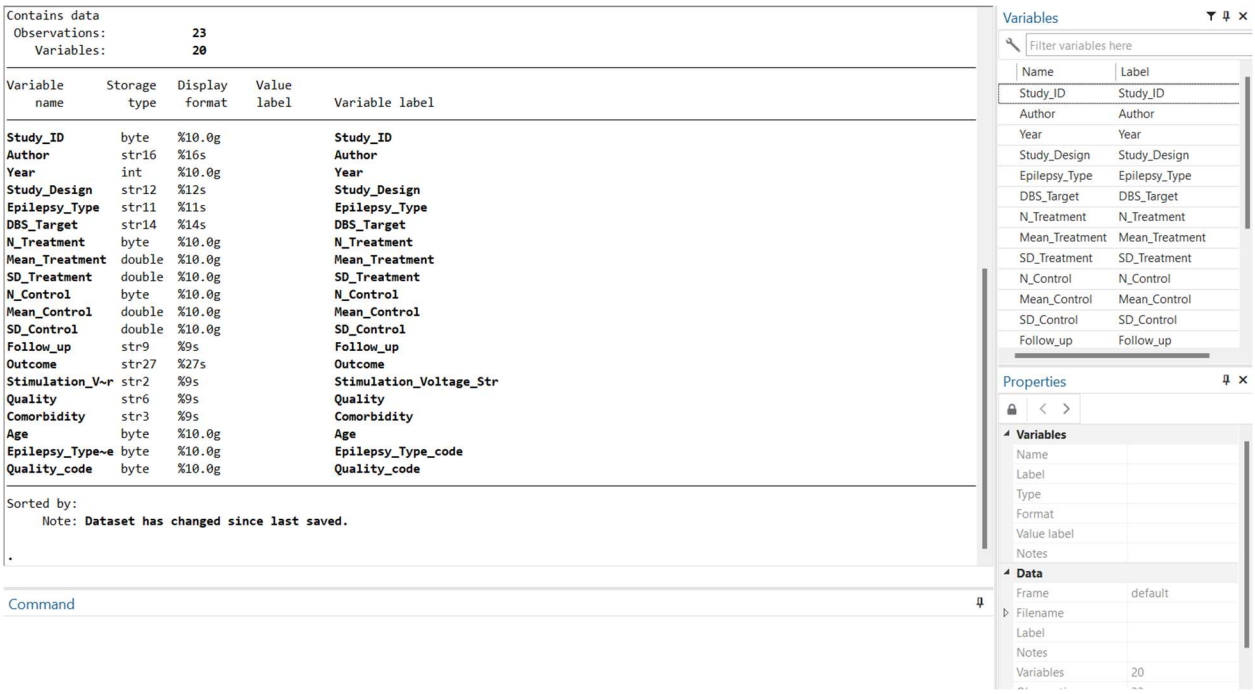
r	Year	Study_Design	Epilepsy_Type	DBS_Target	N_Treatment	Mean_Treatment	SD_Treatment	N_Control	Mean_Control	SD_Control	Follow_up	Outcome	Stimulation_Voltage_Str	Quality	Como
al.	2021	Cohort	Focal	Unilateral ANT	20	69/3	10/9	16	16/7	10/2	24 months	Seizure Frequency Reduction	3V	Low	N
al.	2011	Case-Control	Focal	Bilateral ANT	20	62/4	19/5	40	12/7	15/3	24 months	Seizure Frequency Reduction	6V	Low	Y
et al.	2016	RCT	Focal	Unilateral ANT	21	79/1	9/3	17	14/5	26	24 months	Seizure Frequency Reduction	5V	Low	Y
t al.	2017	RCT	Refractory	Unilateral ANT	39	52/1	17/4	31	19	16/8	12 months	Seizure Frequency Reduction	3V	High	N
al.	2023	Cohort	Focal	Unilateral ANT	54	55/5	11/5	17	31/3	21	6 months	Seizure Frequency Reduction	5V	Medium	N
al.	2021	Case-Control	Generalized	Bilateral ANT	67	8	15/3	39	38/2	14/1	12 months	Seizure Frequency Reduction	4V	High	Y
t al.	2019	Case-Control	Focal	Bilateral ANT	22	46/7	17/1	30	9/5	8/9	12 months	Seizure Frequency Reduction	5V	High	Y
t al.	2023	RCT	Generalized	Bilateral ANT	52	76/6	15	44	11/1	11/2	12 months	Seizure Frequency Reduction	5V	Low	N
. al.	2018	RCT	Generalized	Unilateral ANT	19	47/8	11/9	21	32/1	16/2	24 months	Seizure Frequency Reduction	3V	High	N
et al.	2014	Cohort	Focal	Bilateral ANT	38	66/3	17/4	43	16/9	16/7	24 months	Seizure Frequency Reduction	3V	High	Y
t al.	2018	Cohort	Focal	Unilateral ANT	50	67/1	15/4	57	24/5	16/9	24 months	Seizure Frequency Reduction	4V	Low	N
t al.	2015	RCT	Focal	Unilateral ANT	30	58/2	10/4	37	12/8	14/9	12 months	Seizure Frequency Reduction	5V	Medium	N
. al.	2022	Case-Control	Refractory	Bilateral ANT	49	45/5	16/2	35	34/8	17/2	12 months	Seizure Frequency Reduction	4V	Low	Y
al.	2022	RCT	Refractory	Unilateral ANT	18	74/5	15/6	40	11/6	12/1	12 months	Seizure Frequency Reduction	4V	Medium	N
t al.	2010	RCT	Refractory	Bilateral ANT	33	58/5	11/9	48	33/2	9/6	24 months	Seizure Frequency Reduction	3V	High	Y
et al.	2015	Case-Control	Focal	Unilateral ANT	28	64/3	20	22	22/9	19/8	24 months	Seizure Frequency Reduction	3V	Low	Y
. al.	2022	RCT	Focal	Bilateral ANT	18	69/9	15/1	23	33/6	10/2	6 months	Seizure Frequency Reduction	5V	High	N
et al.	2023	RCT	Focal	Unilateral ANT	52	68/6	16/5	31	23/7	18	24 months	Seizure Frequency Reduction	6V	High	Y
et al.	2018	Case-Control	Refractory	Unilateral ANT	44	69/3	11/7	44	28	10/3	12 months	Seizure Frequency Reduction	4V	Medium	N
al.	2011	Case-Control	Refractory	Bilateral ANT	90	97	12/4	15	6/2	17/3	12 months	Seizure Frequency Reduction	5V	High	N
et al.	2013	Cohort	Refractory	Unilateral ANT	36	59/4	11/6	59	11/6	15/4	6 months	Seizure Frequency Reduction	6V	Medium	Y
al.	2020	RCT	Refractory	Bilateral ANT	36	53/4	20/4	55	25/8	17/2	24 months	Seizure Frequency Reduction	6V	Medium	Y
al.	2018	Case-Control	Refractory	Bilateral ANT	32	78/9	14/5	54	22/8	14	6 months	Seizure Frequency Reduction	5V	High	N

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2.



3.



4.

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```
. summarize N_Treatment Mean_Treatment SD_Treatment N_Control Mean_Control SD_Control
```

Variable	Obs	Mean	Std. dev.	Min	Max
N_Treatment	23	37.73913	17.93984	18	90
Mean_Treat~t	23	62.10435	16.93078	8	97
SD_Treatment	23	14.65217	3.185602	9.3	20.4
N_Control	23	35.56522	13.90496	15	59
Mean_Control	23	21.45652	9.440042	6.2	38.2
SD_Control	23	15.18696	4.122257	8.9	26

```
.
```

5.

```
. tabulate Outcome
```

Outcome	Freq.	Percent	Cum.
Seizure Frequency Reduction	23	100.00	100.00
Total	23	100.00	

6.

```
. list Study_ID Author Year Mean_Treatment Mean_Control in 1/10
```

	Study_ID	Author	Year	Mean_T~t	Mean_C~l
1.	1	Smith et al.	2021	69.3	16.7
2.	2	Jones et al.	2011	62.4	12.7
3.	3	Williams et al.	2016	79.1	14.5
4.	4	Brown et al.	2017	52.1	19
5.	5	Davis et al.	2023	55.5	31.3
6.	6	Miller et al.	2021	8	38.2
7.	7	Wilson et al.	2019	46.7	9.5
8.	8	Moore et al.	2023	76.6	11.1
9.	9	Taylor et al.	2018	47.8	32.1
10.	10	Anderson et al.	2014	66.3	16.9

7.

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```
. genereate MD = Mean_Treatment - Mean_control
command genereate is unrecognized
r(199);

. generate MD = Mean_Treatment - Mean_Control

. list Study_ID Author MD in 1/10
```

	Study_ID	Author	MD
1.	1	Smith et al.	52.6
2.	2	Jones et al.	49.7
3.	3	Williams et al.	64.6
4.	4	Brown et al.	33.1
5.	5	Davis et al.	24.2
6.	6	Miller et al.	-30.2
7.	7	Wilson et al.	37.2
8.	8	Moore et al.	65.5
9.	9	Taylor et al.	15.7
10.	10	Anderson et al.	49.4

.

8.

```
. meta esize N_Treatment Mean_Treatment SD_Treatment N_Control Mean_Control SD_Control, esize(mdiff)

Meta-analysis setting information

Study information
     No. of studies: 23
        Study label: Generic
         Study size: _meta_studysize
       Summary data: N_Treatment Mean_Treatment SD_Treatment N_Control Mean_Control SD_Control

        Effect size
              Type: mdiff
             Label: Mean diff.
          Variable: _meta_es

           Precision
           Std. err.: _meta_se
     Std. err. adj.: None
                 CI: [_meta_cil, _meta_ciu]
           CI level: 95%

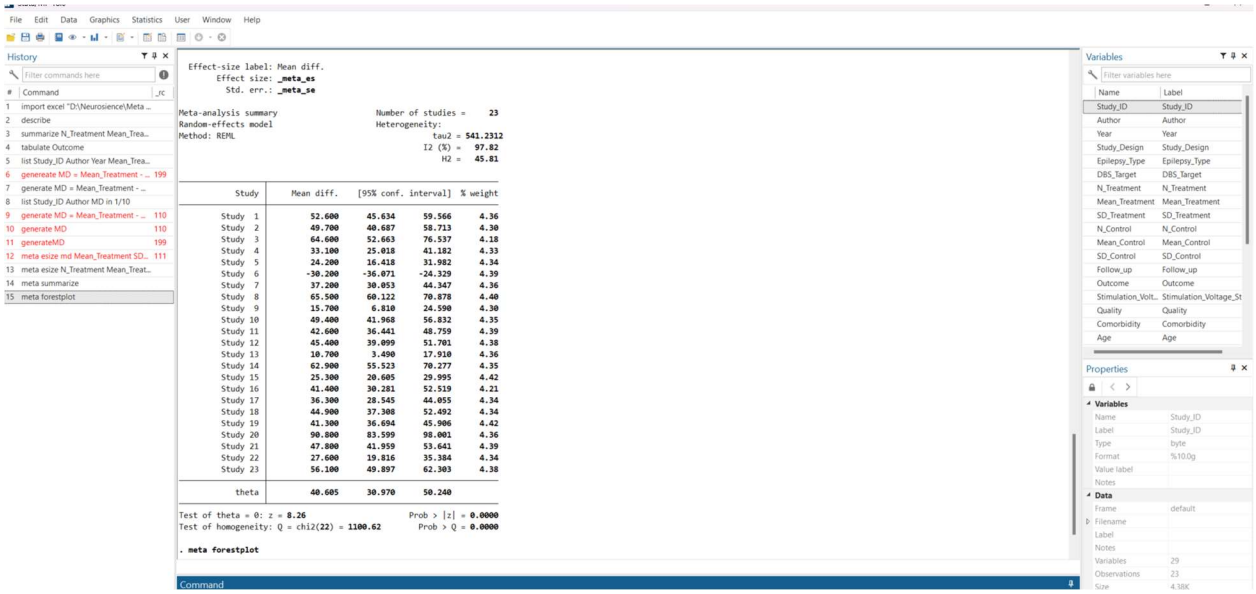
  Model and method
              Model: Random effects
             Method: REML

. meta summarize

  Effect-size label: Mean diff.
        Effect size: _meta_es
          Std. err.: _meta_se
```

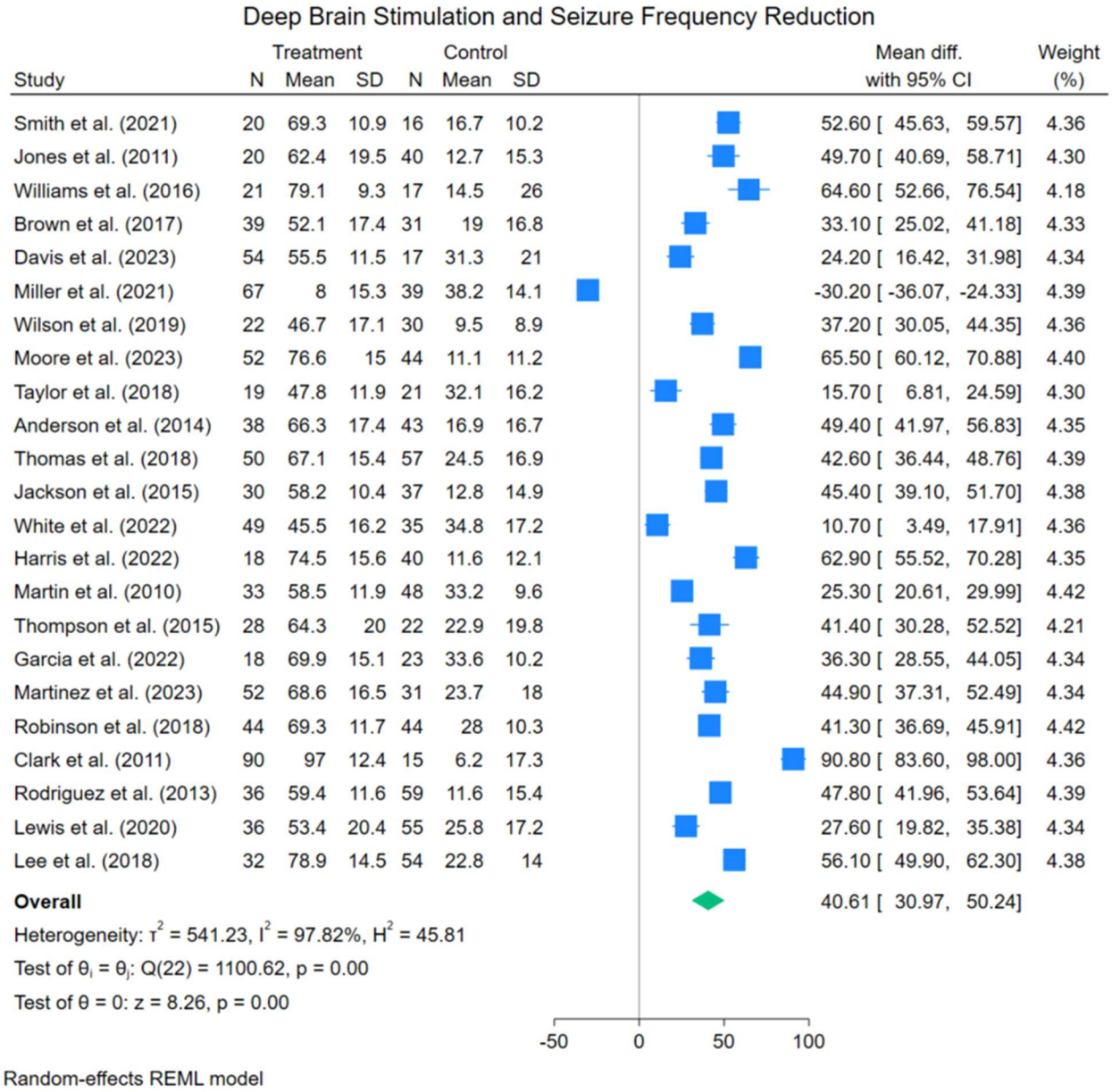
9.

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10.

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11.

1. Overall Effect

The overall effect was statistically significant (MD = 40.61, 95% CI: 30.97–50.24, p < 0.001), indicating that the treatment group had a significantly greater reduction in seizure frequency compared with the control group.

2. Heterogeneity

There was very high heterogeneity among the studies (I^2 = 97.82%), indicating substantial variability in the effect sizes across studies.

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12.

Based on visual inspection of the forest plot, two data points (Studies 6 and 20) appear to be potential outliers.

13.

None of the 23 studies had a non-significant result, as the 95% confidence intervals of all studies did not cross the null value of zero.

14.

The forest plot showed very high heterogeneity among the studies ($I^2 = 97.82\%$, $p < 0.001$), indicating substantial variability in the effect sizes across studies.

15.

Two potential outliers were identified by visual inspection of the forest plot (Study 6 and Study 20), as their effect sizes were substantially different from those of the other studies.

16.

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Notes:

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1. Unicode is supported; see help unicode_advice.
2. More than 2 billion observations are allowed; see help obs_advice.
3. Maximum number of variables is set to 5,000 but can be increased;
   see help set_maxvar.

. import excel "D:\Neurosience\Meta Analysis\2-Meta Analysis\S3_2\Task\Metaanalysis_cleaned.xlsx", sheet("Data_Cleaned") firstrow

(20 vars, 23 obs)

. describe

Contains data
 Observations: 23
 Variables: 20

Variable name	Storage type	Display format	Value label	Variable label
Study_ID	byte	%10.0g		Study_ID
Author	str16	%16s		Author
Year	int	%10.0g		Year
Study_Design	str12	%12s		Study_Design
Epilepsy_Type	str11	%11s		Epilepsy_Type
DBS_Target	str14	%14s		DBS_Target
N_Treatment	byte	%10.0g		N_Treatment
Mean_Treatment	double	%10.0g		Mean_Treatment
SD_Treatment	double	%10.0g		SD_Treatment
N_Control	byte	%10.0g		N_Control
Mean_Control	double	%10.0g		Mean_Control
SD_Control	double	%10.0g		SD_Control
Follow_up	str9	%9s		Follow_up
Outcome	str27	%27s		Outcome
Stimulation_V~r	str2	%9s		Stimulation_Voltage_Str
Quality	str6	%9s		Quality
Comorbidity	str3	%9s		Comorbidity

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Age byte %10.0g Age

Epilepsy_Type~e byte %10.0g Epilepsy_Type_code

Quality_code byte %10.0g Quality_code

Sorted by:

Note: Dataset has changed since last saved.

. summarize N_Treatment Mean_Treatment SD_Treatment N_Control Mean_Control SD_Control

Variable	Obs	Mean	Std. dev.	Min	Max
N_Treatment	23	37.73913	17.93984	18	90
Mean_Treat~t	23	62.10435	16.93078	8	97
SD_Treatment	23	14.65217	3.185602	9.3	20.4
N_Control	23	35.56522	13.90496	15	59
Mean_Control	23	21.45652	9.440042	6.2	38.2
SD_Control	23	15.18696	4.122257	8.9	26

. tabulate Outcome

Outcome	Freq.	Percent	Cum.
Seizure Frequency Reduction	23	100.00	100.00
Total	23	100.00	

. list Study_ID Author Year Mean_Treatment Mean_Control in 1/10

	Study_ID	Author	Year	Mean_T~t	Mean_C~l
1.	1	Smith et al.	2021	69.3	16.7
2.	2	Jones et al.	2011	62.4	12.7

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```
 3. |       3   Williams et al.   2016      79.1      14.5 |
 4. |       4     Brown et al.   2017      52.1        19 |
 5. |       5     Davis et al.   2023      55.5      31.3 |
    |-----------------------------------------------------|
 6. |       6    Miller et al.   2021         8      38.2 |
 7. |       7    Wilson et al.   2019      46.7       9.5 |
 8. |       8     Moore et al.   2023      76.6      11.1 |
 9. |       9    Taylor et al.   2018      47.8      32.1 |
10. |      10   Anderson et al.   2014      66.3      16.9 |
    +-----------------------------------------------------+

. genereate MD = Mean_Treatment - Mean_control
command genereate is unrecognized
r(199);

. generate MD = Mean_Treatment - Mean_Control

. list Study_ID Author MD in 1/10

    +-------------------------------------+
    | Study_ID          Author      MD |
    |-------------------------------------|
 1. |       1     Smith et al.    52.6 |
 2. |       2     Jones et al.    49.7 |
 3. |       3   Williams et al.    64.6 |
 4. |       4     Brown et al.    33.1 |
 5. |       5     Davis et al.    24.2 |
    |-------------------------------------|
 6. |       6    Miller et al.   -30.2 |
 7. |       7    Wilson et al.    37.2 |
 8. |       8     Moore et al.    65.5 |
 9. |       9    Taylor et al.    15.7 |
10. |      10   Anderson et al.    49.4 |
    +-------------------------------------+
```

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. generate MD = Mean_Treatment - Mean_Control

variable MD already defined

r(110);

. generate MD

variable MD already defined

r(110);

. generateMD

command generateMD is unrecognized

r(199);

. meta esize md Mean_Treatment SD_Treatment N_Treatment Mean_Control SD_Control N_Control

variable md not found

r(111);

. meta esize N_Treatment Mean_Treatment SD_Treatment N_Control Mean_Control SD_Control, esize(mdiff)

Meta-analysis setting information

Study information

No. of studies: 23

Study label: Generic

Study size: _meta_studysize

Summary data: N_Treatment Mean_Treatment SD_Treatment N_Control Mean_Control SD_Control

Effect size

Type: mdiff

Label: Mean diff.

Variable: _meta_es

Precision

Std. err.: _meta_se

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Std. err. adj.: None

CI: [_meta_cil, _meta_ciu]

CI level: 95%

Model and method

Model: Random effects

Method: REML

. meta summarize

Effect-size label: Mean diff.

Effect size: _meta_es

Std. err.: _meta_se

Meta-analysis summary Number of studies = 23

Random-effects model Heterogeneity:

Method: REML tau2 = 541.2312

I2 (%) = 97.82

H2 = 45.81

Study	Mean diff.	[95% conf.	interval]	% weight
Study 1	52.600	45.634	59.566	4.36
Study 2	49.700	40.687	58.713	4.30
Study 3	64.600	52.663	76.537	4.18
Study 4	33.100	25.018	41.182	4.33
Study 5	24.200	16.418	31.982	4.34
Study 6	-30.200	-36.071	-24.329	4.39
Study 7	37.200	30.053	44.347	4.36
Study 8	65.500	60.122	70.878	4.40
Study 9	15.700	6.810	24.590	4.30
Study 10	49.400	41.968	56.832	4.35
Study 11	42.600	36.441	48.759	4.39

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```
      Study 12 |      45.400    39.099    51.701    4.38
      Study 13 |      10.700     3.490    17.910    4.36
      Study 14 |      62.900    55.523    70.277    4.35
      Study 15 |      25.300    20.605    29.995    4.42
      Study 16 |      41.400    30.281    52.519    4.21
      Study 17 |      36.300    28.545    44.055    4.34
      Study 18 |      44.900    37.308    52.492    4.34
      Study 19 |      41.300    36.694    45.906    4.42
      Study 20 |      90.800    83.599    98.001    4.36
      Study 21 |      47.800    41.959    53.641    4.39
      Study 22 |      27.600    19.816    35.384    4.34
      Study 23 |      56.100    49.897    62.303    4.38
-----------------+---------------------------------------------------
         theta |      40.605    30.970    50.240
------------------------------------------------------------------------------
Test of theta = 0: z = 8.26                  Prob > |z| = 0.0000
Test of homogeneity: Q = chi2(22) = 1100.62       Prob > Q = 0.0000


. meta forestplot


  Effect-size label: Mean diff.
        Effect size: _meta_es
          Std. err.: _meta_se


. generate Study_Label = Author + " (" + string(Year) + ")"


. list Study_ID Study_Label in 1/10


     +-----------------------------------+
     | Study_ID          Study_Label |
     |-----------------------------------|
  1. |        1     Smith et al. (2021) |
  2. |        2     Jones et al. (2011) |
  3. |        3   Williams et al. (2016) |
```

```
 4. |      4     Brown et al. (2017) |
 5. |      5     Davis et al. (2023) |
    |-----------------------------------|
 6. |      6    Miller et al. (2021) |
 7. |      7    Wilson et al. (2019) |
 8. |      8     Moore et al. (2023) |
 9. |      9    Taylor et al. (2018) |
10. |     10   Anderson et al. (2014) |
    +-----------------------------------+
```

```
. meta forestplot, studylabel(Study_Label)
option studylabel() not allowed
r(198);
```

```
. help meta forestplot
```

```
. meta update, studylabel(Study_Label)
-> meta esize N_Treatment Mean_Treatment SD_Treatment N_Control Mean_Control SD_Control , esize(mdiff) studylabel(Study_Label)
```

```
Meta-analysis setting information from meta esize

 Study information
    No. of studies: 23
       Study label: Study_Label
        Study size: _meta_studysize
      Summary data: N_Treatment Mean_Treatment SD_Treatment N_Control Mean_Control SD_Control

       Effect size
             Type: mdiff
            Label: Mean diff.
         Variable: _meta_es

        Precision
        Std. err.: _meta_se
```

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```
  Std. err. adj.: None
           CI: [_meta_cil, _meta_ciu]
     CI level: 95%

Model and method
        Model: Random effects
       Method: REML

. meta forestplot

Effect-size label: Mean diff.
     Effect size: _meta_es
      Std. err.: _meta_se
     Study label: Study_Label

. meta forestplot, title("Deep Brain Stimulation and Seizure Frequency Reduction")

Effect-size label: Mean diff.
     Effect size: _meta_es
      Std. err.: _meta_se
     Study label: Study_Label

. meta forestplot, nullrefline

Effect-size label: Mean diff.
     Effect size: _meta_es
      Std. err.: _meta_se
     Study label: Study_Label

. . meta forestplot, title("Deep Brain Stimulation and Seizure Frequency Reduction")

Effect-size label: Mean diff.
     Effect size: _meta_es
      Std. err.: _meta_se
```

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```
      Study label: Study_Label


. meta forestplot, nullrefline, title("Deep Brain Stimulation and Seizure Frequency Reduction")
invalid 'title'
r(198);

. meta forestplot, title("Deep Brain Stimulation and Seizure Frequency Reduction") nullrefline

  Effect-size label: Mean diff.
      Effect size: _meta_es
        Std. err.: _meta_se
      Study label: Study_Label

.
```